Sai Sivaram Vangavolu

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SUMMARY

AI Engineer with **3+ years** of experience designing, building, and deploying **Generative AI**, **Large Language Model (LLM)**, and **Machine Learning** applications in **financial services** and enterprise environments, including **Morgan Stanley**. Specialized in **Agentic AI**, **multi-agent orchestration** with **LangGraph** and **LangChain**, **Retrieval-Augmented Generation (RAG)**, **hybrid search**, and **LLM evaluation** with **LangSmith**, **RAGAS**, and **DeepEval**. Builds **production-grade** document intelligence pipelines, **FastAPI** inference services, **AI guardrails**, and **LLMOps/MLOps** workflows on **AWS**, **Azure**, and **Google Cloud Platform (GCP)**. Track record of improving response relevance by **40%**, cutting manual review effort by **65%**, and reducing inference costs by **22%** for **2,000+ users**.

TECHNICAL SKILLS

Programming Languages: Python, SQL, TypeScript, Java
Backend Development: FastAPI, AsyncIO, Pydantic, SQLAlchemy, REST APIs, GraphQL, Microservices, Event-Driven Architecture
Generative AI & LLMs: Large Language Models (LLMs), OpenAI GPT, Anthropic Claude, Google Gemini, Llama, Mistral, Azure OpenAI, Amazon Bedrock, Google Vertex AI, SAP Generative AI Hub, Hugging Face Transformers
Agentic AI: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, DSPy, Model Context Protocol (MCP), Tool Calling, Function Calling, Multi-Agent Systems, Agent Memory, Human-in-the-Loop
Retrieval-Augmented Generation (RAG): Agentic RAG, GraphRAG, Hybrid Search, BM25, Semantic Search, Vector Search, Embeddings, Reranking, Chunking, Pinecone, pgvector, Weaviate, FAISS, Qdrant, Elasticsearch, Neo4j, SAP HANA Cloud Vector Engine
Prompt Engineering: Context Engineering, Prompt Templating, Grounding, Structured Outputs, Chain-of-Thought Prompting, Prompt Versioning
LLM Evaluation & Observability: LangSmith, Langfuse, Arize Phoenix, RAGAS, DeepEval, LLM-as-a-Judge, Golden Datasets, Regression Testing, A/B Testing, OpenTelemetry, Prometheus, Grafana
Responsible AI & Guardrails: NeMo Guardrails, Guardrails AI, Hallucination Detection, PII Redaction, Data Masking, Content Filtering (SAP Orchestration Service), Prompt Injection Defense, AI Governance, Model Risk Management
Fine-Tuning & Model Serving: LoRA, QLoRA, PEFT, vLLM, Ollama, Quantization, Semantic Caching, LiteLLM, Model Routing, Inference Optimization
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Natural Language Processing (NLP), Deep Learning, Anomaly Detection, Fraud Detection, Time-Series Forecasting, Feature Engineering
MLOps & Data Engineering: MLflow, Weights & Biases, Apache Airflow, Apache Kafka, PySpark, Databricks, Snowflake, Docker, Kubernetes, CI/CD, GitHub Actions, Model Deployment, Model Monitoring, Drift Detection
Cloud Platforms & AI Infrastructure: Amazon Web Services (AWS), Amazon SageMaker, AWS Lambda, Amazon EKS, Microsoft Azure, Google Cloud Platform (GCP), SAP Business Technology Platform (SAP BTP), SAP AI Core
Databases & Tools: PostgreSQL, Redis, MongoDB, DynamoDB, Celery, Git, Linux, Pytest, Agile/Scrum

EXPERIENCE

AI Engineer | Morgan Stanley, USA — Nov 2025 – Present

- Designed and deployed production Retrieval-Augmented Generation (RAG) applications for financial research and enterprise knowledge workflows using Python, LangChain, Azure OpenAI, and vector search, with iterative prompt engineering on curated evaluation sets, improving response relevance by 40% for 2,000+ internal users.
- Engineered a hybrid search retrieval pipeline over 500K+ enterprise documents combining BM25 keyword search, dense embeddings, cross-encoder reranking, and optimized chunking, increasing retrieval precision and grounding LLM responses in accurate financial context.
- Built stateful agentic AI workflows in LangGraph with tool calling, persistent agent memory, and human-in-the-loop approval checkpoints to orchestrate multi-step research and document review, eliminating manual hand-offs between analysts.
- Developed end-to-end document intelligence pipelines using OCR, Natural Language Processing (NLP), and LLMs to extract, classify, and summarize documents, with Pydantic structured outputs and schema validation, reducing manual review effort by 65%.
- Built asynchronous LLM inference APIs using FastAPI, AsyncIO, SQLAlchemy, Apache Kafka, Celery, and Redis for event-driven processing, serving 10K+ daily requests at under 2-second p95 latency.
- Established an automated LLM evaluation framework with LangSmith, RAGAS, and DeepEval, measuring faithfulness, answer relevancy, and context precision on golden datasets and gating releases with regression evals in CI/CD, detecting hallucination risks before production.
- Implemented AI guardrails for PII redaction, prompt injection detection, and hallucination detection through input and output validation layers, producing audit-ready evidence for AI governance and model risk management reviews in a regulated banking environment.
- Optimized LLM inference with semantic caching, request batching, and model routing of simpler queries to smaller models, reducing compute costs by 22% with no measurable quality loss.

Founder, R&D (Part-Time, Self-Employed) | Lakshyaa.ai (Remote) — Jul 2026 – Present

- Architected a six-agent decision-support system on Google Vertex AI with Gemini, using a supervisor agent to coordinate four specialist agents and a risk-assessment agent for multi-agent orchestration, delivering ranked cash recommendations at about $0.07 per query.
- Built a Python cash-forecasting and plan-search engine that generates, scores, and ranks candidate plans against business constraints, evaluating 5,000 plans in 1.3 seconds or less and passing 38 of 39 audited test cases.
- Designed a hybrid LLM and deterministic architecture, using Gemini for intent interpretation and strict numerical parsers to map spreadsheet data into 31 input variables and 47 KPIs, eliminating LLM arithmetic errors and keeping financial outputs auditable.
- Built an eval-driven development framework with a 40-seed benchmark, a 20-question agent evaluation set, trace-level review, and 340 automated test files, and applied context engineering to cut agent context by 94%, lowering token cost and latency.

AI Engineer | Accenture, India — May 2022 – Jul 2024

- Built semantic search pipelines by generating embeddings, indexing them in vector databases, and serving ranked results through REST APIs, improving retrieval accuracy by 35% and reducing search latency by 30%.
- Fine-tuned BERT and DistilBERT transformer models with Hugging Face Transformers and PyTorch for document classification and information extraction, combined with Scikit-learn data validation, reducing manual review effort by 45%.
- Developed fraud detection and anomaly detection models using XGBoost, Scikit-learn, feature engineering, and threshold tuning, improving risk identification by 28% and reducing false-positive alerts by 20%.
- Engineered scalable ETL and ML data pipelines with PySpark and Apache Airflow, processing 15M+ records monthly into clean, model-ready datasets for analytics, reporting, and model training.
- Automated model deployment by containerizing models with Docker, deploying to Kubernetes on AWS, and standardizing CI/CD release pipelines, shortening release cycles from days to hours.
- Implemented MLOps workflows with MLflow experiment tracking, model versioning, and Airflow-orchestrated retraining on AWS, making model updates reproducible and reducing update cycles by 35%.
- Developed Python REST APIs and ML inference services with input validation and error handling, enabling downstream enterprise applications to consume real-time model predictions.

EDUCATION

Master of Science in Business Analytics | *University of New Haven, West Haven, CT* — **Dec 2025**
Bachelor of Engineering in Computer Science | *Jawaharlal Nehru Technological University, Kakinada, India* — **May 2023**

CERTIFICATIONS

- SAP Certified Associate: SAP Generative AI Developer
- SAP Certified Associate: Data Analyst – SAP Analytics Cloud
- Google Data Analytics Professional Certificate